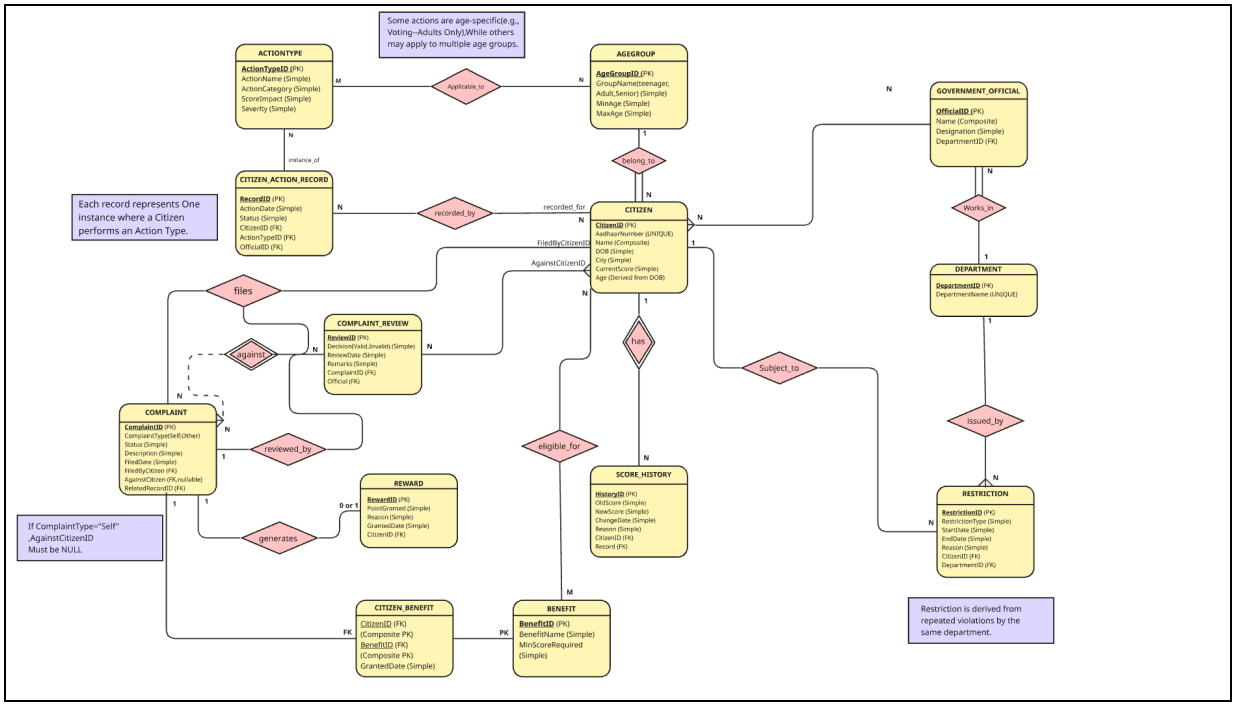
DBMS Project Task - 2

Project Group Number - 47
Tutorial Group Number - 4
Tutorial Room Number - C210
Assigned TAs - Raghav Verma & Rahul Pardasani

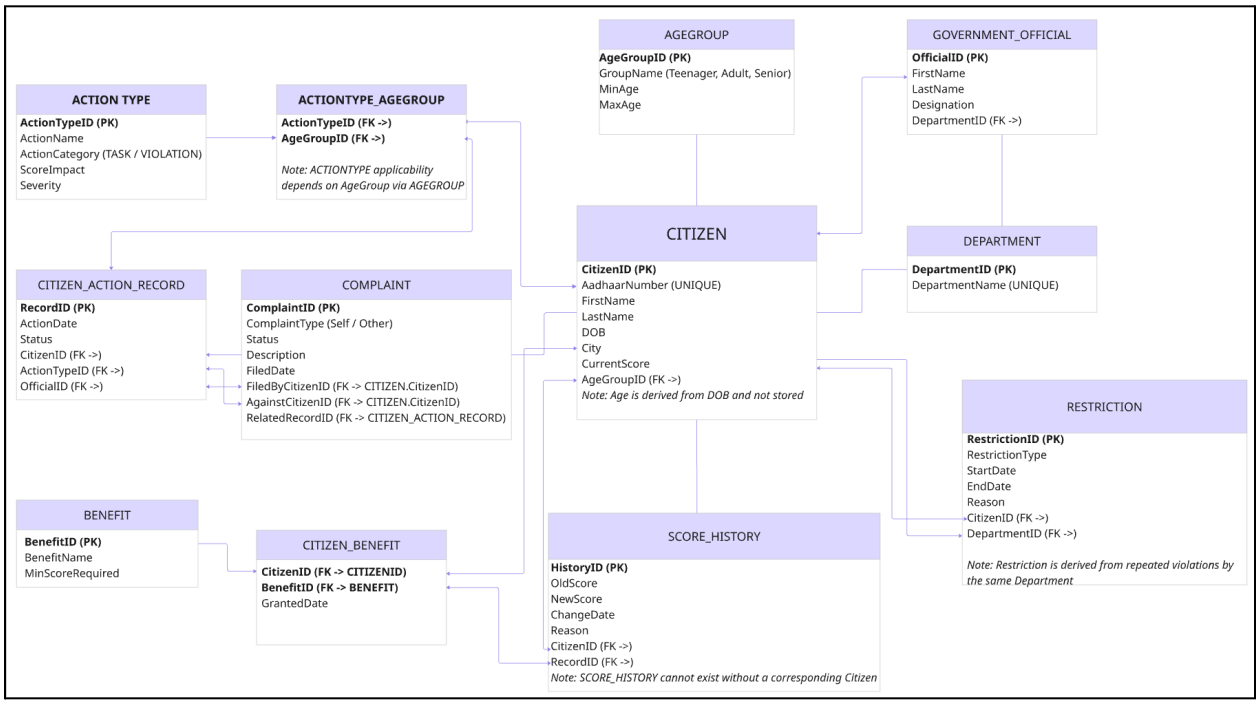
1) Mrityunjai Poddar	2024358
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Citizen Social Responsibility & Trust Score Management System

ER-Diagram [Miro_Link]



Relational Schema [Miro_Link]



Entity Sets Explanation

01. **Citizen -** Central entity of the database, represents individuals whose social actions, score, benefits, complaints and restrictions are tracked in the system.
02. **Age Group -** Categorise citizens age wise, Used to filter actions specific to certain age groups.
03. **Action Type -** Possible Types and categories of actions which exist in the system.
04. **Citizen Action Record -** Stores actual instances where citizens perform an action. This is an associative entity.
05. **Government Official** - Record citizen actions and review complaints.
06. **Department** - Represents different govt departments.
07. **Reward** - Rewards Granted based on correct complaints
08. **Score History** - Logs of changes in score.
09. **Benefit** - Definition of possible benefits a citizen can receive based on its score retention.
10. **Citizen Benefit** - Associative Entity that records which citizen got what benefit in actual.
11. **Restrictions** - Issued by departments to citizens based on repetitive violations.

Relationships & Cardinality Explanation

01. **Citizens belong to AgeGroups (1 : N)** - One age group can contain many citizens, but each citizen belongs to exactly one age group.
02. **ActionType applicable to AgeGroups (M : N)** - One action type can apply to multiple age groups, and one age group can have multiple applicable action types.
03. **Citizen has CitizenActionRecord (1 : N)** - One citizen can have many action records, but each action record is associated with exactly one citizen.
04. **GovernmentOfficial records CitizenActionRecord (1 : N)** - One government official can record many action records, but each action record is recorded by exactly one official.

5. **Citizen files Complaint (1 : N)** - One citizen can file many complaints, but each complaint is filed by exactly one citizen.
6. **Complaint against Citizen (1 : N)** - One citizen can have many complaints filed against them, but each complaint is against exactly one citizen.
7. **Complaint reviewed by ComplaintReview (1 : N)** - One complaint can have multiple reviews, but each complaint review corresponds to exactly one complaint.
8. **GovernmentOfficial reviews ComplaintReview (1 : N)** - One government official can review many complaint reviews, but each complaint review is performed by exactly one official.
9. **Complaint generates Reward (1 : 0/1)** - One complaint may generate at most one reward, and some complaints may not generate any reward.
10. **Citizen has ScoreHistory (1 : N)** - One citizen can have many score history records, but each score history record belongs to exactly one citizen.
11. **Citizen eligible for Benefit (M : N)** - One citizen can be eligible for multiple benefits, and one benefit can be granted to multiple citizens.
12. **Department works in Government Official (1 : N)** - One department can have many government officials, but each government official works in exactly one department.
13. **Department issued by Restriction (1 : N)** - One department can issue many restrictions, but each restriction is issued by exactly one department.
14. **Citizen subject to Restriction (1 : N)** - One citizen can be subject to many restrictions, but each restriction applies to exactly one citizen.